# Scalability & Future Plan

Wandr AI – AI Travel Itinerary Planner

## 1. Current Scalability Posture

| Dimension | Current State | Bottleneck |
|---|---|---|
| Concurrency | FastAPI async ASGI; multiple requests handled concurrently | Groq API rate limit on free tier |
| Database | Supabase PostgreSQL with indexed queries + RLS | Supabase connection pool on free tier |
| Frontend | Static React SPA on Vercel CDN — globally distributed | None for static assets |
| AI throughput | 10 AI requests/user/hour (configurable env var) | Groq API token limits |
| Storage | JSONB itinerary in single trips table | Very large itineraries (>14 days) may slow queries |

## 2. Scaling Path

| Upgrade | Action | Benefit |
|---|---|---|
| Remove cold starts | Upgrade Render to Starter ($7/mo) | Always-on backend |
| Increase DB connections | Upgrade Supabase to Pro ($25/mo) | Pgbouncer connection pooling |
| Increase AI throughput | Upgrade Groq to paid tier | Higher RPM and token limits |
| Caching | Add Redis for repeated similar itineraries | Reduce Groq API calls |
| CDN for API | Add Cloudflare in front of Render | Global latency reduction |

## 3. Planned Future Features (v2+)

| Feature | Description | Effort |
|---|---|---|
| Multi-language output | Let user select itinerary language | Medium |
| Trip sharing | Share a read-only itinerary link with others | Medium |
| Collaborative editing | Multiple users edit a trip in real time | High |
| Flight/hotel price hints | Query flight and hotel APIs for price ranges | High |

| | | |
|---|---|---|
| Offline PWA mode | Cache itinerary in service worker for offline view | Medium |
| AI trip comparison | Generate 2 itinerary variants and let user pick | Medium |
| Mobile app | React Native wrapper around existing API | High |
| Usage analytics | Track which features are used most (privacy-first) | Low |

## 4. Monetisation Options (Future)

- **Freemium:** 3 free trips/month; paid plan for unlimited generation.
- **API as a service:** Expose the itinerary generation API to third-party travel apps.
- **Affiliate revenue:** Link to verified hotel/flight booking platforms (with disclosure).